WORKED SOLUTION

# STEP I 1996, Question 10

Mechanics • Conservation of Momentum

**STEP I 1996, Question 10**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s1-q10

## The key idea

There are two conservation statements to use: momentum is conserved (it was zero before the explosion), and the kinetic energy afterwards is $E$. Together they fix both speeds once the split of the mass is known, so the relative speed becomes a function of how the mass is shared. Minimising it then comes down to a single inequality, and AM–GM is the natural tool because $m_1 + m_2$ is fixed.

## Solution

We model the two parts as particles, assume no mass is lost in the explosion, so $m_1 + m_2 = M$, and assume no external forces act during it.




**Momentum.** The spaceship starts at rest, so the total momentum is zero before and therefore after the explosion. The two momenta $m_1\mathbf{v}_1$ and $m_2\mathbf{v}_2$ must cancel, so the parts move in exactly opposite directions along a line, and

$$m_1 v_1 = m_2 v_2 = p,$$

say, where $p > 0$ is the common magnitude of momentum. Because the parts move in opposite directions, their relative speed really is $v_1 + v_2$.

**Energy.** All of the kinetic energy afterwards comes from the explosion, so

$$E = \tfrac{1}{2}m_1 v_1^2 + \tfrac{1}{2}m_2 v_2^2 = \frac{p^2}{2m_1} + \frac{p^2}{2m_2} = \frac{p^2(m_1+m_2)}{2m_1 m_2} = \frac{p^2 M}{2m_1 m_2}.$$

Hence $p^2 = \dfrac{2Em_1m_2}{M}$.

**Relative speed.** Using $v_1 = p/m_1$ and $v_2 = p/m_2$,

$$v_1 + v_2 = p\left(\frac{1}{m_1} + \frac{1}{m_2}\right) = \frac{pM}{m_1m_2},$$

so

$$(v_1 + v_2)^2 = \frac{p^2M^2}{m_1^2m_2^2} = \frac{2Em_1m_2}{M} \cdot \frac{M^2}{m_1^2m_2^2} = \frac{2EM}{m_1m_2}.$$

> **Ask yourself** $E$ and $M$ are fixed. What is the only thing left that we are free to vary, and which way do we want it to go?

Only the split of the mass can change, and to make $v_1 + v_2$ as small as possible we need the product $m_1m_2$ as large as possible, subject to $m_1 + m_2 = M$. By the AM–GM inequality (both masses are positive),

$$m_1m_2 \le \left(\frac{m_1 + m_2}{2}\right)^2 = \frac{M^2}{4},$$

with equality exactly when $m_1 = m_2 = \frac{1}{2}M$. (Equivalently, $m_1m_2 = \frac{1}{4}M^2 - \frac{1}{4}(m_1 - m_2)^2$.) Therefore

$$(v_1 + v_2)^2 = \frac{2EM}{m_1m_2} \ge \frac{2EM}{M^2/4} = \frac{8E}{M},$$

and since speeds are non-negative we may take square roots:

$$v_1 + v_2 \ge \left(\frac{8E}{M}\right)^{1/2}.$$

The bound is attained when the spaceship splits into two equal halves, each then moving at speed $\sqrt{2E/M}$. So the minimum possible relative speed is $(8E/M)^{1/2}$, as required.

## Checks

- Equal halves: each has kinetic energy $\frac{1}{2}E = \frac{1}{2} \cdot \frac{1}{2}Mv^2$, so $v = \sqrt{2E/M}$ and $2v = \sqrt{8E/M}$, agreeing with the bound.
- Extreme split: as $m_1 \to 0$, $m_1m_2 \to 0$ and the relative speed blows up, which makes sense, because a tiny fragment carrying a share of the energy must move very fast.
- Units: $E/M$ has units of $\mathrm{m^2\,s^{-2}}$, so its square root is a speed.
- Numerically, with $M = 7$ and $E = 3$, scanning $m_1$ over $(0, 7)$ gives a least relative speed of $1.85164\ldots$ at $m_1 = 3.5$, matching $\sqrt{24/7} = 1.85164\ldots$.

## Common slips

- Forgetting to say why the parts move in opposite directions; that is what makes the relative speed $v_1 + v_2$ rather than $|v_1 - v_2|$.
- Minimising over $m_1$ without using $m_1 + m_2 = M$; the constraint is what makes the problem one-dimensional.
- Showing only that $(8E/M)^{1/2}$ is a lower bound, without noting it is actually attained (by $m_1 = m_2$).

## Beyond STEP

This is the simplest example of working in the centre-of-mass frame: in that frame a two-body break-up always produces equal and opposite momenta, and the kinetic energy is $p^2/(2\mu)$, where $\mu = m_1 m_2/M$ is the *reduced mass*. Here $(v_1 + v_2)^2 = 2E/\mu$, and the reduced mass is largest when the masses are equal. The same idea underlies nuclear decays and two-body orbits; Kleppner and Kolenkow, *An Introduction to Mechanics*, develops it carefully.

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